Teaching Assistant (TA) System

Implementation Plan — Full Specification

Project	PhysioGuidance eLearning Platform
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Developer	Gamma Developers
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Standard	NELC Compliant — Health Sector
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■ **Update Notice:** This plan includes an addition to the original spec — TAs can now **view assessment submissions** (read-only, no grading). See Part 4d and the Permission Matrix for details.

1. CONTEXT & CURRENT STATE

The TA role supports instructors during live sessions with participant management, attendance monitoring, and session assistance. This document is the full implementation specification to be handed directly to the development team.

What Already Exists (Backend ~95% Done)

- ✔ TEACHING_ASSISTANT role in RBAC enum (backends/src/database/schema/roles.ts)
- ✔ session_assistants table with complete schema
- ✔ Full service: assign, remove, list, available TAs (session-assistants.service.ts)
- ✔ Controller with 4 endpoints (session-assistants.controller.ts)
- ✔ Frontend API client (frontend/src/lib/api.ts — sessionAssistantsApi)
- ✔ Email notification template (ta-assignment.template.ts)
- ✔ NELC audit events: ASSISTANT_ALLOCATED, ASSISTANT_DEALLOCATED
- ✔ NELC thresholds: ≤25 students = 0 TAs | 26–35 = 1 TA | >35 = 2 TAs

What Is Missing (All Frontend + Some Backend)

- ✗ No TA dashboard (no route /dashboard/ta)
- ✗ No TA assignment UI anywhere in instructor or manager dashboards
- ✗ No course-level TA assignment — only session-level currently exists
- ✗ Instructor cannot suggest a TA for manager approval
- ✗ TA cannot join Zoom sessions or view attendance
- ✗ TA cannot view assessment submissions (NEW requirement)
- ✗ No TA management section in admin or manager dashboards

2. ARCHITECTURE DECISIONS

Decision	Choice	Rationale
TA Assignment Level	Course-level + auto-propagate to sessions	Assigning per-session is tedious for 20+ sessions. Course-level auto-applies to all sessions with per-session override.
Assignment Authority	Manager assigns; Instructor suggests	Final approval must remain with the Course Manager per client requirements.
TA Dashboard	Lightweight read-only dashboard	TAs need: assigned sessions list, attendance view, assessment submissions view, Zoom join. No course ownership, no editing.
Auto-Allocation	Stays DISABLED	Already disabled per client request. System shows NELC threshold warnings only.

3. DATABASE CHANGES

3a. New Table: course_assistants

Course-level TA assignment that auto-propagates to sessions.

```
course_assistants
■■■ id: UUID (PK)
■■■ courseId: UUID → courses.id (CASCADE DELETE)
■■■ userId: UUID → users.id
■■■ status: 'suggested' | 'approved' | 'rejected' (default: 'approved')
■■■ suggestedBy: UUID → users.id (nullable — set when instructor suggests)
■■■ approvedBy: UUID → users.id (nullable — set when manager approves)
■■■ assignedAt: TIMESTAMP (default: now)
■■■ approvedAt: TIMESTAMP (nullable)
■■■ UNIQUE(courseId, userId)
```

3b. Schema Index Update

File: backends/src/database/schema/index.ts — Export the new course_assistants table.

4. BACKEND — COURSE ASSISTANTS MODULE

4a. New Module Files

File	Purpose
backends/src/modules/course-assistants/course-assistants.module.ts	NestJS module registration
backends/src/modules/course-assistants/course-assistants.service.ts	Business logic
backends/src/modules/course-assistants/course-assistants.controller.ts	API endpoints
backends/src/modules/course-assistants/dto/assign-course-ta.dto.ts	Validation DTO

4b. Service Methods

Method	Caller	Logic
assignTA(courseId, userId, assignedBy)	Manager / Admin	status='approved'; auto-propagates to all existing sessions via session-assistants.service
suggestTA(courseId, userId, suggestedBy)	Instructor	status='suggested'; sends in-app notification to all managers
approveTA(courseId, userId, approvedBy)	Manager / Admin	Changes status to 'approved'; propagates to sessions; emails TA
rejectTA(courseId, userId)	Manager / Admin	Changes status to 'rejected'; notifies instructor
removeTA(courseId, userId)	Manager / Admin	Deletes from course_assistants + all session_assistants rows
getCourseTAs(courseId)	Any Authorized	Returns list with user info + status badges
propagateToSession(courseId, sessionId)	Internal	Auto-assigns all approved course TAs to a newly created session

4c. API Endpoints

Method	Endpoint	Allowed Roles
GET	/courses/:courseId/assistants	INSTRUCTOR, MANAGER, ADMIN
POST	/courses/:courseId/assistants	MANAGER, ADMIN (direct assign)
POST	/courses/:courseId/assistants/suggest	INSTRUCTOR (suggest only)
PATCH	/courses/:courseId/assistants/:userId/approve	MANAGER, ADMIN
PATCH	/courses/:courseId/assistants/:userId/reject	MANAGER, ADMIN
DELETE	/courses/:courseId/assistants/:userId	MANAGER, ADMIN

4d. Hook Into Session Creation

File: backends/src/modules/sessions/sessions.service.ts

After a session is created (approx. line ~170), call courseAssistantsService.propagateToSession(courseId, newSessionId) to auto-assign all approved course-level TAs to the new session.

4e. TA Session List Endpoint

File: backends/src/modules/sessions/sessions.service.ts

Add method getAssignedSessions(userId) — returns all sessions where the user appears in session_assistants. Powers the TA dashboard sessions view.

Controller: add GET /sessions/my-assigned guarded by TEACHING_ASSISTANT role.

4f. TA Zoom Access

TAs join as participants (join_url), not as hosts. The session detail endpoint already returns zoomLink. Ensure TA can fetch session details for their assigned sessions only — no changes to the start-url endpoint required.

4g. TA Attendance View

Existing endpoint GET /enrollments/session/:sessionId/attendance already permits TEACHING_ASSISTANT role. No backend change needed.

NEW — 4h. TA Assessment Submissions View

TAs can view assessment submissions for their assigned courses/sessions in read-only mode. They cannot grade, edit scores, or publish results.

Attribute	Value	Notes
Endpoint	GET /assessments/:assessmentId/submissions	Returns all submissions for an assessment
Guard	TEACHING_ASSISTANT — read only	Must be assigned to the session/course the assessment belongs to
Scope	Assigned courses/sessions only	TA cannot access submissions for courses they are not assigned to
Actions	View only	No grade input, no score override, no feedback submission
Data	Student name, submission content, submitted_at	Scores and grades are HIDDEN unless already published by instructor

Backend file to modify: backends/src/modules/assessments/assessments.controller.ts — Add TEACHING_ASSISTANT to the roles guard on the GET submissions endpoint, with an additional scope check confirming the TA is assigned to the relevant session.

5. FRONTEND — TA ASSIGNMENT UI

5a. Instructor — Course Detail (Suggest TA)

File: frontend/src/components/instructor-dashboard/CourseDetailContent.tsx

- New tab or section: 'Teaching Assistants'
- List current TAs with status badges (green = approved, yellow = pending, red = rejected)
- 'Suggest TA' button with dropdown of available users with TEACHING_ASSISTANT role
- Calls POST /courses/:courseId/assistants/suggest

5b. Instructor — Sessions (TA Deficit Warning)

File: frontend/src/components/instructor-dashboard/SessionsContent.tsx

- Session card shows number of TAs assigned
- Warning badge if below NELC threshold (calculated from enrolled count)

5c. Manager — Course Detail (TA Management)

File: frontend/src/components/course-manager-dashboard/CoursesContent.tsx or new CourseDetailContent.tsx

- Full TA management panel in course detail view
- Direct assign (approved immediately) + approve/reject instructor suggestions
- Remove TA button with confirmation
- NELC compliance indicator: 'X TAs assigned, Y required for Z enrolled students'

5d. Manager — Session Cards (TA Indicators)

File: frontend/src/components/course-manager-dashboard/SessionsContent.tsx

Session cards show TA count and a warning icon if below NELC threshold.

5e. Admin — Role Assignment (No Change)

Admin can already assign TEACHING_ASSISTANT role via existing user management. No frontend changes required.

6. FRONTEND — TA DASHBOARD

6a. Routes & Layout Files

File	Purpose
frontend/src/app/[locale]/(dashboards)/dashboard/ta/page.tsx	TA dashboard entry route
frontend/src/app/[locale]/(dashboards)/dashboard/ta/layout.tsx	TA dashboard layout wrapper
frontend/src/components/ta-dashboard/TADashboardContent.tsx	Home page — today's sessions + quick actions
frontend/src/components/ta-dashboard/TANavbar.tsx	Sidebar navigation (Dashboard, Sessions)
frontend/src/components/ta-dashboard/TASessionsContent.tsx	All assigned sessions list + filters
frontend/src/components/ta-dashboard/TAAttendanceView.tsx	Read-only attendance view per session
frontend/src/components/ta-dashboard/TAAssessmentSubmissionsView.tsx	NEW — Read-only submissions view per assessment

6b. TA Dashboard Home

- Today's assigned sessions with status: upcoming / live / ended
- Quick actions: Join Zoom (live sessions only), View Attendance
- Weekly mini-calendar of assigned sessions
- No course management, no enrollment management

6c. TA Sessions View

- Full list of assigned sessions across all courses
- Filters: date, course, status
- Each session card: title, course name, date/time, type, status
- Live sessions → 'Join Zoom' + 'View Attendance'
- Ended sessions → 'View Attendance' (read-only)
- Scheduled sessions → info display only

6d. TA Attendance View

Read-only view of student attendance for a session:

- Student list with present / absent / late status
- Zoom join and leave timestamps
- No marking or editing capability (view only)
- Per-session filtering by name or student ID

NEW — 6e. TA Assessment Submissions View

A read-only view of student submissions for assessments linked to assigned sessions/courses.

- Accessible from the Sessions view: each session card shows a 'View Submissions' button if an assessment is linked
- Lists all student submissions: student name, student ID, submission date/time, submission content (essay, file, quiz answers)
- Scores and grades are HIDDEN unless the instructor has already published results

- No grade input field, no score field, no feedback entry — all editing controls are absent from the DOM
- Export is disabled for TAs (only instructors and managers can export gradebooks)
- File: frontend/src/components/ta-dashboard/TAAssessmentSubmissionsView.tsx

6f. TA Navigation Sidebar

Two items only: Dashboard (home) and Sessions. No courses, no enrollments, no content management.

6g. Role-Based Redirect

File: frontend/src/app/[locale]/(dashboards)/dashboard/page.tsx

When a user with activeRole = TEACHING_ASSISTANT hits /dashboard, redirect to /dashboard/ta.

7. FRONTEND API EXTENSIONS (API.TS)

courseAssistantsApi

Method	Endpoint
getCourseTAs(courseId)	GET /courses/:courseId/assistants
assignTA(courseId, data)	POST /courses/:courseId/assistants
suggestTA(courseId, data)	POST /courses/:courseId/assistants/suggest
approveTA(courseId, userId)	PATCH /courses/:courseId/assistants/:userId/approve
rejectTA(courseId, userId)	PATCH /courses/:courseId/assistants/:userId/reject
removeTA(courseId, userId)	DELETE /courses/:courseId/assistants/:userId

taApi

Method	Endpoint
getMyAssignedSessions()	GET /sessions/my-assigned
getSessionSubmissions(sessionId, assessmentId)	GET /assessments/:assessmentId/submissions (NEW)

8. NOTIFICATIONS

Trigger	Recipients	Message
Instructor suggests TA	In-app to all managers	"Instructor X suggested TA Y for Course Z"
Manager approves suggestion	In-app to instructor + Email to TA	"Your TA suggestion for [name] was approved"
Manager rejects suggestion	In-app to instructor	"Your TA suggestion for [name] was rejected"
Manager directly assigns TA	Email to TA (existing template) + In-app to instructor	"TA [name] has been assigned to your course"

9. PERMISSION MATRIX

Action	TA	Instructor	Manager	Admin
View assigned sessions	Own only	Own course	All	All
Join Zoom (as participant)	Assigned	Own (host)	—	—
View attendance	Assigned	Own sessions	All	All
View assessment submissions	Assigned (read-only)	Own sessions	All	All
Mark attendance	No	Yes	Yes	Yes
Grade assessments	No	Yes	Yes	Yes
Suggest TA	No	Own courses	—	—
Assign TA (direct)	No	No	Yes	Yes
Approve/reject suggestion	No	No	Yes	Yes
Remove TA	No	No	Yes	Yes
Create/edit courses	No	Yes	Yes	Yes
Create/edit sessions	No	Yes	Yes	Yes
Manage enrollments	No	Yes	Yes	Yes
Export gradebook	No	Yes	Yes	Yes

10. IMPLEMENTATION ORDER (PHASES)

Phase	Scope	Files
Phase 1	DB schema + migration	course-assistants.ts, schema/index.ts, migration file
Phase 2	Course assistants backend module	course-assistants.module.ts, .service.ts, .controller.ts, DTO
Phase 3	Hook into session creation	sessions.service.ts — call propagateToSession() after create
Phase 4	TA session list + my-assigned endpoint	sessions.service.ts, sessions.controller.ts
Phase 5	Assessment submissions endpoint for TA	assessments.controller.ts — add TA read-only guard (NEW)
Phase 6	Manager UI — assign/approve TAs	CoursesContent.tsx or new CourseDetailContent.tsx
Phase 7	Instructor UI — suggest TAs	CourseDetailContent.tsx, SessionsContent.tsx
Phase 8	TA Dashboard — Home + Sessions + Attendance	New route + TADashboardContent, TASessionsContent, TAAttendanceView
Phase 9	TA Dashboard — Assessment Submissions view (NEW)	TAAssessmentSubmissionsView.tsx + taApi extension
Phase 10	Role redirect + translations	dashboard/page.tsx, messages/en.json, messages/ar.json

11. FILES TO CREATE

File	Purpose
backends/src/database/schema/course-assistants.ts	Course-level TA assignment table
backends/src/modules/course-assistants/course-assistants.module.ts	NestJS module
backends/src/modules/course-assistants/course-assistants.service.ts	Business logic
backends/src/modules/course-assistants/course-assistants.controller.ts	API endpoints
backends/src/modules/course-assistants/dto/assign-course-ta.dto.ts	Validation DTO
frontend/src/app/[locale]/(dashboards)/dashboard/ta/page.tsx	TA dashboard route
frontend/src/app/[locale]/(dashboards)/dashboard/ta/layout.tsx	TA dashboard layout
frontend/src/components/ta-dashboard/TADashboardContent.tsx	TA home page
frontend/src/components/ta-dashboard/TANavbar.tsx	TA sidebar navigation
frontend/src/components/ta-dashboard/TASessionsContent.tsx	Sessions list view
frontend/src/components/ta-dashboard/TAAttendanceView.tsx	Read-only attendance view
frontend/src/components/ta-dashboard/TAAssessmentSubmissionsView.tsx	NEW — Read-only submissions view

12. FILES TO MODIFY

File	Purpose
backends/src/database/schema/index.ts	Export course_assistants
backends/src/modules/sessions/sessions.service.ts	Propagate TAs on session create; add getAssignedSessions()
backends/src/modules/sessions/sessions.controller.ts	Add GET /sessions/my-assigned for TA role
backends/src/modules/assessments/assessments.controller.ts	NEW — Add TA read-only guard on GET submissions endpoint
backends/src/app.module.ts	Register CourseAssistantsModule
frontend/src/lib/api.ts	Add courseAssistantsApi and taApi (incl. getSessionSubmissions)
frontend/src/components/instructor-dashboard/CourseDetailContent.tsx	Add TA suggestion section
frontend/src/components/instructor-dashboard/SessionsContent.tsx	Add TA deficit warning badge
frontend/src/components/course-manager-dashboard/CoursesContent.tsx	Add TA assignment management section
frontend/src/components/course-manager-dashboard/SessionsContent.tsx	Add TA count indicators on session cards
frontend/src/app/[locale]/(dashboards)/dashboard/page.tsx	Add TA role redirect to /dashboard/ta
frontend/messages/en.json	TA-related English translations
frontend/messages/ar.json	TA-related Arabic translations

13. NELC COMPLIANCE CHECKLIST

- ✔ TA is a separate role — not student, not instructor
- ✔ TA does not consume a student seat (separate table, not in enrollments)
- ✔ TA assignment controlled by institution — Manager assigns, not auto
- ✔ Instructor can suggest TA; Manager must approve before activation
- ✔ No automatic assignment without institutional control
- ✔ NELC thresholds enforced: ≤25=0 TA | 26-35=1 TA | >35=2 TAs (warnings shown)
- ✔ All assignments audit-logged (ASSISTANT_ALLOCATED / ASSISTANT_DEALLOCATED)
- ✔ TA has limited permissions: view only, no course ownership, no grading
- ✔ Assessment submission view is read-only — no grade capability exposed (NEW)
- ✔ TA Zoom access is join_url (participant) not start_url (host)

14. VERIFICATION STEPS (QA)

1. Admin creates user with TEACHING_ASSISTANT role
2. Manager assigns TA to a course → TA auto-assigned to all existing sessions
3. Instructor creates new session → TA auto-assigned via propagateToSession()
4. Instructor suggests a different TA → Managers receive in-app notification
5. Manager approves suggestion → TA assigned, receives assignment email
6. TA logs in → redirected to /dashboard/ta, sees assigned sessions
7. TA clicks 'Join Zoom' on a live session → joins as participant (not host)
8. TA views attendance → read-only student list, no edit controls visible
9. TA views assessment submissions → student submissions listed, no grade fields present, scores hidden if unpublished
10. TA attempts to access instructor routes → receives 403 Forbidden
11. TA cannot export gradebook → export button is absent from the TA submissions view
12. NELC audit log shows ASSISTANT_ALLOCATED / ASSISTANT_DEALLOCATED entries correctly

This document was prepared by Gamma Developers for the PhysioGuidance eLearning Platform. All implementation must comply with NELC Health Sector eLearning Standards, SCFHS/CPD requirements, WCAG 2.1 AA accessibility, and Saudi PDP Data Protection Law.